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END SEMESTER EXAMINATION - 2025

Name of the Course: M. Tech (Electric Vehicle Technology) Semester: 1st
Name of the Paper: Automotive Technology Paper Code: MEV 103
Time: 3 Hours Maximum Marks: 100

Note:-

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1	(20marks)	**CO1**
(a)	List the main components of an automobile and discuss their functions.	
(b)	What are the various drive systems used in automobiles? Briefly explain Front-Wheel Drive (FWD), and Rear-Wheel Drive (RWD) with neat sketches.	
(c)	Define the following I.C. engine terms: Bore, Stroke, Top Dead Centre (TDC), Bottom Dead Centre (BDC), Clearance Volume, Swept Volume, and Compression Ratio.	
Q2	(20 marks)	**CO2**
(a)	Differentiate between Single Point Fuel Injection (SPFI) and Multi Point Fuel Injection (MPFI) systems. Write their advantages and disadvantages with suitable examples	
(b)	Write short notes on the Common Rail Direct Injection (CRDI) system and its advant over conventional diesel injection systems.	
(c)	Compare the carburetor-based fuel supply system with a fuel injection system in terms of performance, fuel economy, and emission characteristics.	
Q3	(20 marks)	**CO3**
(a)	What are the main elements of a transmission system, provide a schematic diagram and describe the functions of each element.	
(b)	Explain the functions and operation of a torque converter in an automatic transmission.	
(c)	Describe the layout and operation of a sliding mesh gearbox with the help of a neat sketch	
Q4	(20 marks)	**CO4 CO5**
(a)	Explain the requirements of a steering system in an automobile. Explain steering mechanism and its main elements with a neat schematic diagram.	
(b)	Describe the construction and operation of an Anti-lock Braking System (ABS). List its major components and explain its advantages over conventional brakes.	
(c)	Explain Hydraulic Power Steering, highlighting their components and advantages.	
Q5	(20 marks)	**CO6**
(a)	Discuss the advantages and limitations of using CNG and LNG as automobile fuels.	
(b)	What are the main advantages and limitations of using hydrogen as an automobile fuel? Explain its production methods briefly.	
(c)	What are the benefits and challenges of using biodiesel blends such as B20 in conventional diesel engines?	